Unit: Unit 3: Functions
Topic: Definition of Relation and Function, Domain, Range, and Notations

Name: ________________
Date: ____________

Advanced (Habanero)

Q1. Which of the following is a function?
(A) $\{(1,2),(2,3),(3,2)\}$
(B) $\{(1,2),(1,3),(2,3)\}$
(C) $\{(1,1),(2,2),(3,3)\}$
(D) $\{(1,1),(1,2),(2,2)\}$
(E) $\{(1,2),(2,1),(3,3)\}$

Q2. What is the domain of $f(x)=\sqrt{x}$?
(A) $(-\infty,\infty)$
(B) $(-\infty,0]$
(C) $[0,\infty)$
(D) $(-\infty,0)$
(E) $(0,\infty)$

Q3. The relation $R=\{(x,y)|x^2+y^2=4\}$ is a
(A) function
(B) relation, but not a function
(C) neither a relation nor a function
(D) one-to-one function
(E) onto function

Q4. What is the range of $f(x)=|x|$?
(A) $(-\infty,0]$
(B) $[0,\infty)$
(C) $(-\infty,0)$
(D) $(-\infty,\infty)$
(E) $[1,\infty)$

Q5. The graph of a function $f(x)$ passes the vertical line test if
(A) no vertical line intersects the graph in more than one point
(B) every vertical line intersects the graph in exactly one point
(C) no horizontal line intersects the graph in more than one point
(D) every horizontal line intersects the graph in exactly one point
(E) the graph is a straight line

Q6. If $f(x)=2x+1$, then $f(2)$ is equal to
(A) 2
(B) 3
(C) 4
(D) 5
(E) 6

Q7. The equation $y=|x-2|$ represents a
(A) linear function
(B) quadratic function
(C) absolute value function
(D) exponential function
(E) logarithmic function

Q8. The domain of $f(x)=\frac{1}{x}$ is
(A) $(-\infty,0]$
(B) $[0,\infty)$
(C) $(-\infty,0)\cup(0,\infty)$
(D) $(-\infty,\infty)$
(E) $[1,\infty)$

Open Ended Questions (FRQ)

Q9. Determine the domain and range of the function $f(x)=\sqrt{x^2-4}$ and explain why.

Q10. Consider the relation $R=\{(x,y)|x=|y|\}$. Determine whether this relation is a function and explain why.

Chef's Tips

- Tip 1: Use the vertical line test to determine if a relation is a function.
- Tip 2: When finding the domain of a function, consider any restrictions on the input values.
- Tip 3: When finding the range of a function, consider any restrictions on the output values.